

Lab 10 Solutions(Grp 1-8)

Atomic detail :

```
class AtomDetail:
    def __init__(self, e, p, n):
        self.e = e
        self.p = p
        self.n = n
    def massNum(self):
        return self.p+ self.n

    def atomNum(self):
        return self.p

    def isNeutral(self):
        if self.e == self.p:
            return True
        else:
            return False
```

```
e, p, n = [int(x) for x in input().split()]
```

```
m = AtomDetail(e,p,n)
```

```
print(m.massNum())
print(m.atomNum())
print(m.isNeutral())
```

Rectangle ! :

```
class Rectangle:
    def __init__(self, A,B,C,D):
        self.A = A
        self.B = B
        self.C = C
        self.D = D
    def getSides(self):
        Sides = [0,0]
```

```

        Sides[0] = abs(self.A[0]-self.B[0])
        Sides[1] = abs(self.A[1]-self.C[1])
        return Sides
    def getArea(self,Sides):
        return Sides[0]*Sides[1]
    def getPerimeter(self,Sides):
        return 2*(Sides[0]+Sides[1])
a = [int(x) for x in input().split()]
b = [int(x) for x in input().split()]
c = [int(x) for x in input().split()]
d = [int(x) for x in input().split()]
A = Rectangle(a,b,c,d)
Sides = A.getSides()
print(A.getPerimeter(Sides))
print(A.getArea(Sides))

```

Savings :

```

class Worker:
    def __init__(self, A, B):
        self.A = A
        self.B = B

    def savings(self):
        return (B/100)*A

```

```

A, B = [float(x) for x in input().split()]

```

```

obj = Worker(A,B)

```

```

save = obj.savings()

```

```

print(f'{save:.1f}')

```

Corona virus :

```

class covid:
    def __init__(self, idx, strength):
        self.idx = idx

```

```
self.strength = strength
```

```
class human:
```

```
    def __init__(self, idx, immu):
```

```
        self.idx = idx
```

```
        self.immu = immu
```

```
    def check(self, st):
```

```
        if self.immu>st:
```

```
            return False
```

```
        else:
```

```
            return True
```

```
n = int(input())
```

```
lv = []
```

```
for x in range(n):
```

```
    idx, st = [int(x) for x in input().split()]
```

```
    obj = covid(idx,st)
```

```
    lv.append(obj)
```

```
m = int(input())
```

```
lh = []
```

```
for x in range(m):
```

```
    idx, st = [int(x) for x in input().split()]
```

```
    obj = human(idx,st)
```

```
    lh.append(obj)
```

```
for x in range(m):
```

```
    for y in range(n):
```

```
        tv = lh[x].check(lv[y].strength)
```

```
        tp = '('+str(lh[x].idx)+' '+str(lv[y].idx)+''
```

```
        print(tp, tv)
```

Complex arithmetic :

```
class Complex:
```

```
    def __init__(self,a:int, b:int):
```

```
    self.a = a
    self.b = b
def getSum(self, c):
    a = self.a + c.a
    b = self.b + c.b
    return a,b
def getProduct(self, c):
    return self.a*c.a - self.b*c.b ,self.a*c.b + self.b*c.a
```

```
li = []
for i in range(2):
    a = list(map(int, input().split()))
    li.append(a)
c = Complex(li[0][0], li[0][1])
d = Complex(li[1][0], li[1][1])
```

```
for i in c.getSum(d):
    print(i, end=" ")
print()
for i in c.getProduct(d):
    print(i, end=" ")
```